CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)
DEVANG PATEL INSTITUTE OF ADVANCE TECHNOLOGY & RESEARCH (DEPSTAR)

PROJECT PROPOSAL SUBMISSION

Semester:	**5th**	**Project ID: PRJ_CSE_5_2026_28**

Details of Project Team

Student IDs: **24DCS145(TL) , 24DCS135 , 24DCS140**

Sr. No	**Parameters**	
A.	**Project Title**	**RapidDoc:** AI-Powered Document Intelligence & Editing Platform
B.	**Domain of Project Definition**	**Artificial Intelligence (AI), Natural Language Processing (NLP), Document Engineering, Web Application Development**
C.	**Project Objectives and Scope:** Project objectives are specific, measurable goals that the project aims to achieve. They provide a clear direction and serve as benchmarks for evaluating the project's success. **Scope:** The project scope outlines the boundaries and deliverables of the project, detailing what will and will not be included. It sets the context for what the project will accomplish and helps manage stakeholder expectations.	**Objectives:** • Develop a unified web platform that combines traditional document editing with AI-powered document understanding. • Enable users to upload PDF and DOCX documents and perform editing operations (text, headers, footers, fonts, images, logos) through a single interface. • Integrate AI capabilities for translation, summarization, notes generation, MCQ generation, flashcard creation, viva question generation, and keyword extraction. • Implement a natural language command interface so users can instruct the platform in plain English. • Provide real-time document preview and export in editable DOCX format. • Eliminate the need for users to switch between multiple tools such as Microsoft Word, Adobe Acrobat, translators, and summarizers. **Scope:** • RapidDoc covers the complete document lifecycle — upload, edit, AI-process, preview, and export — within a single web application. The platform supports PDF and DOCX input and produces DOCX output. AI features are powered by a third-party AI API or a self-trained NLP model (pending mentor review). The platform targets students, faculty, businesses, and researchers, and does not replace dedicated word processors but enhances them with AI intelligence and natural language interaction.

D.	**Background Study of Existing System** The background study involves an examination of the existing relevant systems or technology that the project seeks to improve or replace. This includes an analysis of its architecture, functionalities, strengths, and weaknesses.	• Current document workflows are highly fragmented. Users rely on Microsoft Word for editing, Adobe Acrobat for PDF viewing and export, online translators for language conversion, and separate summarizer or note-generation apps for AI-assisted study. Switching between these tools causes loss of context, repeated formatting effort, and reduced productivity. • Existing AI document tools such as ChatPDF, Humata, and Adobe AI Assistant provide Q&A capabilities over documents but do not offer direct editing (replacing text, fonts, images, headers, footers). Conversely, traditional editors like Microsoft Word provide editing but lack intelligent AI features such as summarization, MCQ generation, or multilingual translation. • RapidDoc addresses this gap by combining both engines — a document editing engine and an AI intelligence engine — into one unified web platform, so that users can edit and understand a document from a single interface using simple natural language commands.
E.	**Methodology and Approach** **Methodology:** The methodology section outlines the systematic procedures/method/ workflow/flowchart/architecture and techniques that will be used to conduct the project. This section also describes the practical steps and strategies that will be implemented to achieve the project's goals.	The project follows an iterative Agile development methodology with modular implementation. **Technology Stack:** • Frontend: React.js (web interface, document preview, natural language command input) • Backend: FastAPI (Python) for document processing and AI pipeline management • Document Processing: python-docx and PyMuPDF for DOCX and PDF manipulation • AI Engine: Third-party AI API (Gemini / OpenAI) OR self-trained NLP model for summarization, translation, MCQ generation, and keyword extraction (final choice pending mentor review) • Storage: Local / cloud storage for uploaded and processed documents • Export: DOCX as primary output format; PPTX or TXT as secondary (pending finalization) **Workflow:** Upload Document → Enter Natural Language Command → AI Understands User Intent → Document Processing Engine Executes Operations → Preview Updated Document → Download Final Document

F.	**Tentative Project Plan, Timeline and individual role** The project timeline provides the project schedule,	**Phase 1:** Requirement Analysis & Literature Survey **Phase 2:** System Design, Architecture & Database Design **Phase 3:** Document Upload, Parsing & Editing Engine (PDF + DOCX) **Phase 4:** Natural Language Command Interface & Intent Recognition **Phase 5:** AI Pipeline — Summarization, Translation, MCQ, Notes, Keyword Extraction **Phase 6:** Preview, Export & Branding / Logo Replacement Module **Phase 7:** Frontend Integration & UI/UX Refinement **Phase 8:** Testing, Optimization, Deployment & Documentation

G.	**Innovation and Originality** **Innovation:** The innovation aspect of the project refers to the novel features, methodologies, or technologies introduced to address the problem in a new and effective way. **Originality:** Originality emphasizes the project's uniqueness and contribution to the field. It involves the creation of new knowledge, concepts, or products that have not been previously developed.	**Innovation:** DocuMind introduces a unified document intelligence platform that combines a traditional document editing engine with an AI-powered understanding layer. Users can replace text, headers, footers, fonts, images, and logos, and in the same workflow generate summaries, MCQs, study notes, translations, and key insights — all through simple natural language commands. No existing tool offers both editing and intelligence in one interface. **Originality:** Unlike existing AI document tools (ChatPDF, Humata, Adobe AI) that only support Q&A over documents, and unlike traditional editors (Microsoft Word) that lack AI intelligence, DocuMind merges full document editing, AI-powered document understanding and content generation, and a natural language command interface into one unified web platform. This significantly reduces the time and tools required to work with documents while demonstrating applied NLP, document engineering, and full-stack web development skills.